take the entire age of the universe to reach us. This defines the particle horizon, and it's the farthest out we can observe anything at all, even in principle. Knowing that the universe is about 13.8 billion years old, logic would tell you that the particle horizon must be a sphere of radius 13.8 billion light-years. But that's assuming a static universe. In actual fact, since the universe has been expanding all that time, something just close enough to send its light to us 13.8 billion years ago is now much farther away—approximately 45 billion light-years. So we can define the observable universe to be a sphere of about 45 billion light-years in radius, centered on us.[fn4]

The closest we can get to seeing that "edge" is the cosmic microwave background, whose light comes from almost as far as the particle horizon. But a bit closer to us, we can also see ancient galaxies that are now more than 30 billion light-years away. The light we see from those galaxies started traveling through the universe long before they got to such incredible distances, though. If not, we wouldn't be able to see them at all, since the light coming from them now[fn5] can't ever reach us. It turns out that in a uniformly expanding universe, where the more distant things are receding more quickly, it is inevitable that there is a distance beyond which the apparent recession speed is faster than the speed of light, so light can't catch up.

"Wait!" you might be saying. "Nothing can travel faster than light!" This is a fair point, but it doesn't actually lead to a contradiction. While nothing can travel faster than light *through* space, there's no rule that limits how quickly things can happen to find themselves farther apart because they are sitting still in a space that's getting bigger between them.

The distance at which galaxies are currently moving away from us faster than light is surprisingly close, given how far we can actually see. We call it the Hubble radius, and it's around 14 billion light-years from here. I mentioned in Chapter 3 that we can label the distance to objects by their redshift factors—the amount that their light is shifted toward the red (low frequency/long wavelength) part of the spectrum due to the expansion of the universe. An object at the Hubble radius would have a redshift of about 1.5, meaning the light wave, and the universe itself, has stretched out to two and a half times its original length since the light was emitted.[fn6] But even that utterly unimaginable distance is, in cosmological terms, just around the corner. We've seen individual supernovae out to redshifts of almost 4. The